

Generalized Composite Probing in Mandarin

1 Introduction

It is commonly assumed that phrasal movement in natural languages can be strictly classified as either *A-movement* or *Ā-movement*, which are associated with distinct properties: A-movement but not Ā-movement creates new antecedents for anaphor binding and is not subject to weak crossover or Principle C reconstruction; by contrast, Ā-movement but not A-movement is long-distance, crossing c-commanding NPs (i.e., exhibiting the lack of A-minimality/locality effects) and finite CPs (i.e., exhibiting the lack of clause-boundedness) (see e.g., Postal 1971; Chomsky 1977, 1981; a.o. and Richards 2014 for a comprehensive overview of these properties). Contra the *positional* view of the A/Ā-distinction, which holds that the distinct properties associated with A-movement and Ā-movement are derived from the distinct A-positions and Ā-positions that A-movement and Ā-movement target, respectively (see, e.g., Chomsky 1981, 1995; Mahajan 1990; Déprez 1989; Miyagawa 2010), from a *featural* view of the A/Ā-distinction, distinct properties associated with A-movement and Ā-movement are derived from distinct [A]-feature (e.g., [ϕ], [CASE], [D]) and [Ā]-feature (e.g., [WH], [REL], [TOP], [FOC]) which trigger A-movement and Ā-movement, respectively (Van Urk 2015; Longenbaugh 2017; Lohninger, Kovač & Wurmbrand 2022; Lohninger & Yip 2023; Chen 2023, 2025; a.o.). Furthermore, the possibility of *composite probing* allows [A]-feature and [Ā]-feature present on the same head to probe together (via *conjunctive satisfaction*, see Scott 2021), attracting the closest NP with both a matching [A]-feature and a matching [Ā]-feature (Van Urk 2015; Longenbaugh 2017; Lohninger, Kovač & Wurmbrand 2022; Chen 2023, 2025; a.o.). The featural view of the A/Ā-distinction and the possibility of composite probing together predict the existence of *composite A/Ā-movement*, triggered by the composite probe [A+Ā].

The predicted existence of composite A/Ā-movement has two implications: First, composite A/Ā-movement should exhibit properties of both A-movement and Ā-movement under the standard diagnostics. Positive evidence has been found in Dinka movement to Spec, CP (e.g., topicalization and relativization) (Van Urk 2015), English *tough*-movement (Longenbaugh 2017), and Mandarin passivization (Chen 2023, 2025), which are associated with the same mix of A/Ā-properties, as summarized in (1). Second, composite A/Ā-movement should be restricted by the distribution of composite probes. Positive evidence has come from a contrast between Dinka (Van Urk 2015), where composite A/Ā-movement can cross finite CP boundaries, and English and Mandarin, where composite A/Ā-movement can only cross boundaries of non-finite clauses – those lacking a CP projection – but not finite ones (Longenbaugh 2017; Chen 2023, 2025). This

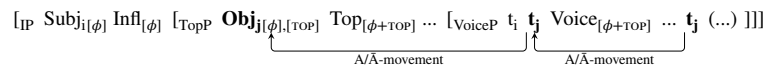
follows from cross-linguistic variations on the distribution of composite probes on the C head and a general ban on improper composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement (see e.g., Longenbaugh 2017; Chen 2023, 2025), which stems from the ban on improper A-movement after $\bar{A}$ -movement (see e.g., May 1979; Chomsky 1981; Abels 2007; Neeleman & Van De Koot 2010; Williams 2011), which in turn stems from the inherent differences between [A]-features and [$\bar{A}$]-features (see e.g., Neeleman & van De Koot 2010; Obata & Epstein 2011; and Chen 2023, 2025 for more detailed discussion). Specifically, it has been proposed that in Dinka, the C head hosts a composite probe [A+ $\bar{A}$] (Van Urk 2015), whereas in English and Mandarin, the C head hosts pure $\bar{A}$ -probes but crucially not the composite probe [A+ $\bar{A}$] (Longenbaugh 2017; Chen 2023, 2025). Consequently, in Dinka, composite A/ $\bar{A}$ -movement can proceed successive-cyclically via Spec, CPs, whereas in English and Mandarin, composite A/ $\bar{A}$ -movement cannot follow from an intermediate step of $\bar{A}$ -movement to Spec, CP (Longenbaugh 2017; Chen 2023, 2025).

This paper aims to expand cross-linguistic evidence for composite probing, with a specific focus on Mandarin. I argue that composite probing by the composite probe [A+ $\bar{A}$] is generally observed in Mandarin, in the sense that multiple heads projected in the low IP area host the composite probe [A+ $\bar{A}$], and that the Voice head, as a phase head, generally hosts the composite probe [A+ $\bar{A}$] for purposes of successive-cyclic movement. The evidence will come from two types of topicalization and focalization in Mandarin, which exhibit mixed A/ $\bar{A}$ -properties, as summarized also in (1). Specifically, I will argue that *IP-internal topicalization and focalization* involve successive-cyclic composite A/ $\bar{A}$ -movement via Spec, VoicePs, which terminate at IP-internal Spec, TopP and Spec, FocP, respectively, as illustrated in (2), while *IP-external topicalization and focalization* can involve intermediate steps of composite A/ $\bar{A}$ -movement to Spec, VoicePs, followed by a terminating step of $\bar{A}$ -movement to IP-external Spec, TopP and Spec, FocP, respectively, as illustrated in (3).

- (1) *Mixed A/ $\bar{A}$ -properties of composite A/ $\bar{A}$ -movement vs. Mandarin topicalization and focalization*

Properties	A	$\bar{A}$	Dinka, English, Mandarin composite A/ $\bar{A}$ -movement	Mandarin IP-internal topicalization focalization	Mandarin IP-external topicalization focalization
New antecedents for anaphor binding	✓	*	✓	✓	*
No weak crossover	✓	*	✓	✓	* / ✓
No reconstruction for Principle C	✓	*	✓	✓	* / ✓
Long-distance ...	*	✓	✓	✓	✓
... crossing c-commanding NPs					✓
Long-distance ...	*	✓	✓(D)	*	✓
... crossing finite CPs			*(E, M)		

- (2) a. *Mandarin IP-internal topicalization as successive-cyclic composite movement*



- b. *Mandarin IP-internal focalization as successive-cyclic composite movement*

[_{IP} Subj_j[_φ] Infl[_φ] [_{FocP} **Obj**_i[_φ],[_{Foc}] Foc[_φ+_{Foc}] ... [_{VoiceP} t_i t_j Voice[_φ+_{Foc}] ... t_j (...)]]]

A/ $\bar{A}$ -movement A/ $\bar{A}$ -movement

- (3) a. *Mandarin IP-external topicalization via intermediate composite movement*

[_{TopP} **Obj**_i[_φ],[_{Top}] Top[_{Top}] [_{IP} Subj_j[_φ] Infl[_φ] ... [_{VoiceP} t_i t_j Voice[_φ+_{Top}] ... t_i (...)]]]

$\bar{A}$ -movement A/ $\bar{A}$ -movement

- b. *Mandarin IP-external focalization via intermediate composite movement*

[_{FocP} **Obj**_i[_φ],[_{Foc}] Foc[_{Foc}] [_{IP} Subj_j[_φ] Infl[_φ] ... [_{VoiceP} t_i t_j Voice[_φ+_{Foc}] ... t_i (...)]]]

$\bar{A}$ -movement A/ $\bar{A}$ -movement

The remainder of the paper is organized as follows: In section 2, I will review the two implications of composite A/ $\bar{A}$ -movement – its association with mixed A/ $\bar{A}$ -properties and its restriction by the distribution of composite probes – drawing on cross-linguistic data from Dinka topicalization and relativization (Van Urk 2015), English *tough*-movement (Longenbaugh 2017), and Mandarin passivization (Chen 2023, 2025). In section 3, I will present the evidence for generalized composite probing in Mandarin from IP-internal and IP-external topicalization and focalization. Specifically, I will argue that their association with mixed A/ $\bar{A}$ -properties and locality restrictions follow directly from composite A/ $\bar{A}$ -movement and the distribution of composite probes on the C and Voice heads. In section 4, I will conclude with a distribution of A-, $\bar{A}$ -, and composite probes in Mandarin low IP area and left periphery.

Throughout the paper, the sources of linguistic examples and judgements are cited when they are based on external references. Uncited examples and judgements are my own and have been checked with 5+ additional native speakers.

2 Two implications of composite A/ $\bar{A}$ -movement

2.1 Mixed A/ $\bar{A}$ -properties

The first implication of composite A/ $\bar{A}$ -movement is that it should exhibit properties of both A-movement and $\bar{A}$ -movement under the standard diagnostics. Positive evidence has been found in Dinka movement to Spec, CP (e.g., topicalization and relativization) (Van Urk 2015), English *tough*-movement (Longenbaugh 2017), and Mandarin passivization (Chen 2023, 2025), which are associated with the same mix of A/ $\bar{A}$ -properties. Specifically, like A-movement and unlike $\bar{A}$ -movement, these movements (i) create new antecedents for anaphor binding, as seen in (4); (ii) are not subject to weak crossover, as seen in (5); and (iii) do not show reconstruction effects for Principle C, as seen in (6).¹

¹Abbreviations in glosses follow the Leipzig glossing conventions, with the following additions: C=complementizer; CL=classifier; CS=construct state; DIST=distributivity marker; EXP=experiential aspect; HAB=habitual; MOD=modification marker; NF=nonfinite; OV=object voice; P=preposition; SV=subject voice.

- (4) *Composite A/ $\bar{A}$ -movement: new antecedents for anaphor binding*
- a. *Dinka topicalization*
 Bòl_i à-cí [DP àkékòl-tí è ròt-dè_i] __ piòolic.
 Bol 3S-PRF.OV story-that P self-SG.3SG criticize.NF
 ‘Bol, that story about himself has criticized (him).’ (Van Urk 2015: 111, ex. 37a)
- b. *English tough-movement*
 [Jon and Mary]_i were hard for each other_i’s friends to get along with ___i.
 (Longenbaugh 2017: ex. 14b; see also Ruys 2000; Pesetsky 2013)
- c. *Mandarin passivization*
 (Lian) Lisi_i (dou) bei (ta-)ziji_i-de tonghuo bipo Zhangsan pai jingcha
 even Lisi DIST BEI 3SG-self’s complice force Zhangsan send police
 zhuazou-le ___i.
 arrest-PRF
 Lit. ‘(Even) Lisi was forced Zhangsan to send the police to arrest by his complice.’ (Chen 2025: ex. 47a)
- (5) *Composite A/ $\bar{A}$ -movement: no weak crossover*
- a. *Dinka topicalization*
 Dhùk ébén_i à-cí thók-dè_i __ kâac.
 boy every 3S-PRF.OV goat-CS-SG.SG bite.NF
 ‘Every boy_i, his_i goat bit (him).’ (Van Urk 2015: 110, ex. 35a)
- b. *English tough-movement*
 No employee_i will be easy for us to get his_i boss to fire ___i. (Longenbaugh 2017: ex. 14a; see also Lasnik & Stowell 1991)
- c. *Mandarin passivization*
 Mei-ge xiaotou_i dou bei ta_i-de tonghuo bipo Zhangsan pai jingcha
 every-CL thief DIST BEI 3SG’s complice force Zhangsan send police
 zhuazou-le ___i.
 arrest-PRF
 Lit. ‘Every thief_i was forced Zhangsan to send the police to arrest (him_i) by his_i complice.’ (Chen 2025: ex. 47b)
- (6) *Composite A/ $\bar{A}$ -movement: no Principle C reconstruction*
- a. *Dinka topicalization*
 [DP Mánh è [Máyèn kù Àyén]_i] cùkè_i __ tùñ.
 brother P Mayen.GEN and Ayen PRF.3PL see.NF
 ‘The brother of [Mayen and Ayen]_i, they_i have seen (him).’ (Van Urk 2015: 114, ex. 43a)
- b. *English tough-movement*
 [Mary_i’s father]_j is tough for her_i to get along with ___j. (Longenbaugh 2017: ex. 14c; see also Mulder & den Dikken 1992; Takahashi 2011)

c. *Mandarin passivization*

(Lian) zhe-ge xiaotou_i-de tonghuo_j (dou) bei ta_i bipo Zhangsan pai
 even this-CL thief's complice DIST BEI 3SG force Zhangsan send
 jingcha zhuazou-le ___j.
 police arrest-PRF

Lit. '(Even) this thief_i's complice was forced Zhangsan to send the police
 to arrest by him_i.' (Chen 2025: ex. 47c)

Unlike A-movement and like $\bar{A}$ -movement, these movements (i) can be long-distance, crossing other c-commanding NPs (i.e., exhibiting the lack of A-minimality/locality effects), as seen in the above examples (4), (5), and (6); and (ii) induce islands for extraction, as seen in (7).²

(7) *Composite A/ $\bar{A}$ -movement: islands for extraction*

a. *Dinka relativization*

*Tòny à-cí Áyèn [DP ràn [CP mër __]] t̃iŋ.
 pot 3S-PRF.OV AyeN.GEN person.CS decorate.SV see.NF
 Lit. 'A pot, AyeN has seen someone who is decorating __.' (Van Urk 2015: 99, ex. 13c)

b. *English tough-movement*

*Where_i was *Syntactic Structures*_j enjoyable to read ___j ___i? (Longen-
 baugh 2017: ex. 15c; see also Chomsky 1977; Rezac 2006)

c. *Mandarin passivization*

??Wo xiang-zhidao Zhangsan_i bei (jingcha) renwei/huaiyi/xiangxin ___i
 1SG want-know Zhangsan BEI police think/suspect/believe
 weishenme mousha Lisi.
 why murder Lisi
 Int. 'I want to know the reason x such that Zhangsan was thought/suspected/believed
 to murder Lisi for x (by the police).' (Chen 2025: ex. 48a)

2.2 Distribution of composite probes

The second implication of composite A/ $\bar{A}$ -movement is that it should be restricted by the distribution of composite probes. Positive evidence has come from a contrast in Dinka, English and Mandarin with respect to the possibility of composite A/ $\bar{A}$ -movement across finite clause boundaries: in Dinka, topicalization and relativization can cross finite CP boundaries, as seen in (8) (Van Urk 2015); in English, *tough*-movement is degraded (but possible) for non-subjects and impossible for subjects across a phasal CP-projection, as seen in (9) (Longenbaugh 2017; see also Postal 1971; Bres-

²In Mandarin, the *wh*-adjunct *weishenme* 'why' undergoes covert movement to its scope position in the matrix Spec, CP and hence is island-sensitive (Huang 1982; Tsai 1994/2014); the ill-formedness of (i-a) indicates that extraction of a non-argument out of a BEI-construction is impossible. See Chen (2023, 2025) for more details.

nan 1972; Chomsky 1973; Lasnik & Fiengo 1974; Browning 1987; Rezac 2006);³ in Mandarin, cross-finite-clause passivization is possible from a finite subject position, an instance of hyper-raising to subject, but impossible from a finite object position, as seen in (10) (Chen 2023, 2025).

(8) *Dinka cross-finite-clause topicalization*

- a. Bòl_i à-cí_i [DP thùrá è ròt-dè_i] nyòth [CP kè cùukù ___
Bol 3S-PRF.OV picture P self-SG.3SG show.NF C PRF.1PL
tùñ].
see.NF
'Bol, a picture of himself has shown that we have seen (him).' (Van Urk 2015: 111, ex. 37b)
- b. Mòc èbèñ_i à-yí_i tiéet-dè_i luéel [CP è ___ thèt].
man every 3S-HAB.OV wife-SG.3SG say.NF C cook.SV
'Every man_i, his_i wife says (he) is cooking.' (Van Urk 2015: 110, ex. 36a)
- c. [DP Máñh è [Máyèn kù Àyén]_i] à-yùukù tàak [CP cùikè_i ___
brother P Mayen.GEN and Ayen 3S-HAB.PL think.NF PRF.3PL
tùñ].
see.NF
'The brother of [Mayen and Ayen]_i, we think they_i have seen (him).' (Van Urk 2015: 114, ex. 44a)

(9) *English cross-finite-clause tough-movement*

- a. ??[*]John_i was easy to show [CP that Bill killed ____i]. (Longenbaugh 2017: ex. 38a; see also Lasnik & Fiengo 1974)
- b. ?[*]Kim_i is tough for me to believe [CP that Sandy would ever marry ____i]. (Longenbaugh 2017: ex. 38b; see also Hukari & Levine 1991)
- c. ?[]Mary_i is tough for me to believe [CP that John would ever marry ____i]. (Longenbaugh 2017: ex. 38c; see also Kaplan & Bresnan 1982)
- d. *Jon_i is hard to believe [CP ____i liked Sue]. (Longenbaugh 2017: ex. 37a)

(10) *Mandarin cross-finite-clause passivization*

- a. *Lisi_i bei (jingcha) renwei/huaiyi/xiangxin [CP Zhangsan hui mousha
Lisi BEI police think/suspect/believe Zhangsan will murder
____i].

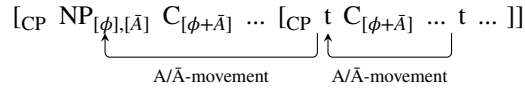
Int. 'Lisi was thought/suspected/believed that Zhangsan will murder (him) (by the police).' (Chen 2025: ex. 23b, based on Ting 1998: ex. 28c)
- b. Zhangsan_i bei (jingcha) renwei/huaiyi/xiangxin [CP ____i hui mousha
Zhangsan BEI police think/suspect/believe will murder
Lisi].
Lisi

³Judgements immediately preceding the square brackets and in the square brackets and are provided by Longenbaugh (2017) and the other cited author(s), respectively.

Lit. ‘Zhangsan was thought/suspected/believed that (he) will murder Lisi (by the police).’ (Chen 2025: ex. 23c, based on Her 2009: ex. 25a)

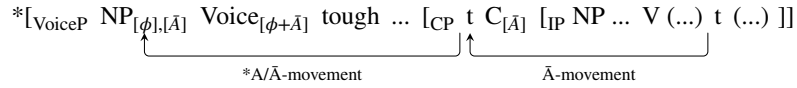
This follows from cross-linguistic variations on the distribution of composite probes on the C head and a general ban on improper composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement (see e.g., Longenbaugh 2017; Chen 2023, 2025), which stems from the ban on improper A-movement after $\bar{A}$ -movement (see e.g., May 1979; Chomsky 1981; Abels 2007; Neeleman & Van De Koot 2010; Williams 2011), which in turn stems from the inherent differences between [A]-features and [$\bar{A}$]-features (see e.g., Neeleman & van De Koot 2010; Obata & Epstein 2011; and Chen 2023, 2025 for more detailed discussion). Specifically, in Dinka, the C head hosts a composite probe [A+ $\bar{A}$], hence Dinka topicalization and relativization can involve intermediate steps of composite A/ $\bar{A}$ -movement to Spec, CPs, as illustrated in (11) (Van Urk 2015).

(11) *Dinka composite A/ $\bar{A}$ -movement via Spec, CP*



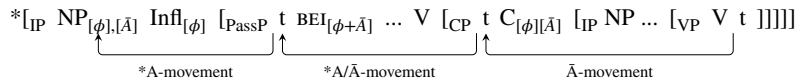
Unlike Dinka, in English and Mandarin, the C head hosts pure $\bar{A}$ -probes but crucially not the composite probe [A+ $\bar{A}$], hence English *tough*-movement from a finite subject position, which involves an intermediate step of $\bar{A}$ -movement to Spec, CP, cannot be followed by further composite A/ $\bar{A}$ -movement, as illustrated in (12); degraded (but possible) instances of English *tough*-movement from finite non-subject positions are derived without an intermediate step of $\bar{A}$ -movement to Spec, CP (Longenbaugh 2017).

(12) *No improper composite-after- $\bar{A}$ -movement in English tough-movement*

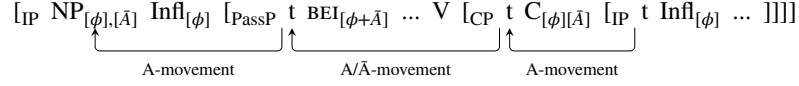


In Mandarin, passivization from a finite object position involves an intermediate step of $\bar{A}$ -movement to Spec, CP, which cannot be followed by further composite A/ $\bar{A}$ -movement, as illustrated in (13). Furthermore, in Mandarin, the C head also hosts a pure A-probe, which is active in the case of hyper-raising to subject, hence, Mandarin passivization from a finite subject position involves an intermediate step of A-movement to Spec, CP, triggered by the pure A-probe on the C head, which can be followed by further composite A/ $\bar{A}$ -movement, as illustrated in (14). Importantly, in Mandarin, despite the presence of both [A]-feature and [$\bar{A}$]-features on the C head, they probe separately and independently (Chen 2023, 2025).

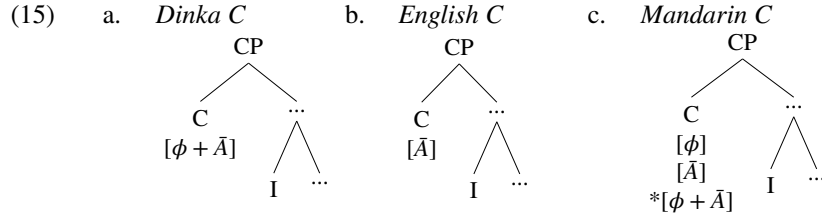
(13) *No improper composite-after- $\bar{A}$ -movement in Mandarin passivization*



(14) *Hyper-raising to subject in Mandarin passivization*



The three-way difference in the feature composition of the probe on the C head across the three languages are illustrated in (15) (Chen 2023, 2025; see also Lohninger, Kovač & Wurmbrand 2022).



It is commonly assumed that the Voice head is also a phase head (see e.g., Chomsky 1986 et seq.). In Dinka, composite A/ $\bar{A}$ -movement is generally observed, suggesting that the Voice head generally hosts the composite probe [A+ $\bar{A}$] for purposes of successive-cyclic movement (Van Urk 2015). In English, only *tough*-movement involves composite A/ $\bar{A}$ -movement, suggesting that only the Voice head(s) involved in the path of *tough*-movement can host the composite probe [A+ $\bar{A}$] (contra Longenbaugh 2017; but see Chen 2023, 2025). In Mandarin, in addition to passivization, topicalization and focalization also exhibit mixed A/ $\bar{A}$ -properties. In section 2, I will argue that in Mandarin, multiple heads projected in the low IP area host the composite probe [A+ $\bar{A}$], and that the Voice head, as a phase head, generally hosts the composite probe [A+ $\bar{A}$] for purposes of successive-cyclic movement.

3 Two types of topic and focus movement in Mandarin

3.1 Movement into the low IP area and left periphery

In Mandarin, topicalization and focalization can target either an IP-internal or IP-external position, either below or above the grammatical subject in Spec, IP (see e.g., Qu 1994; Shyu 1995; Ting 1995; Paul 2002, 2005; Kuo 2009). In (16-a) and (16-b), the IP-internal or IP-external topic or focus NP is linked to a gap in the post-verbal object position.^{4,5}

⁴In the focalization constructions, *lian* ‘even’ is a focus marker and *dou* is a predicate quantifier, according to Shyu (1995). The syntax and semantics of *dou* have been a topic of much discussion but they are not the focus of this paper; for an overview of the distribution and interpretations of *dou*, see e.g., Lin (1998); Xiang (2008); a.o.

⁵It has been noted that a fronted NP must receive a contrastive topic or focus reading when it surfaces in an IP-internal position (see e.g., Qu 1994; Shyu 1995; Ting 1995; Paul 2002, 2005; Kuo 2009). Such a requirement extends to instances of VP and CP topicalization and focalization, which are discussed in section 3.4.

- (16) a. *IP-internal topicalization/focalization*
 [_{IP} Lisi [(lian) **zhe-jian shi**]_i (dou) diaocha-le _____i].
 Lisi even this-CL matter DIST investigate-PRF
 ‘Lisi, (even) this matter_i, investigated (it_i).’
 b. *IP-external topicalization/focalization*
 [(Lian) **zhe-jian shi**]_i [_{IP} Lisi (dou) diaocha-le _____i].
 even this-CL matter Lisi DIST investigate-PRF
 ‘(Even) this matter_i, Lisi investigated (it_i).’

In (17), the IP-internal or IP-external topic or focus NP is linked to an object gap inside a complex NP island; the ill-formedness of (17) suggests that the dependency between the topic or focus NP and the gap is established via movement.

- (17) a. *IP-internal topicalization and focalization: Island-sensitive*
 *Jingcha [(lian) **Lisi**]_i (dou) zhuazou-le [_{NP} na-ge _____j da-le _____i de
 police even Lisi DIST arrest-PRF that-CL hit-PRF REL
 ren_j].
 person
 INT: ‘The police, (even) Lisi_i, arrested that person who hit (him_i).’
 b. *IP-external topicalization and focalization: Island-sensitive*
 *[(Lian) **Lisi**]_i, jingcha (dou) zhuazou-le [_{NP} na-ge _____j da-le _____i de
 even Lisi police DIST arrest-PRF that-CL hit-PRF REL
 ren_j].
 person
 INT: ‘(Even) Lisi_i, the police arrested that person who hit (him_i).’

Furthermore, in IP-internal topicalization and focalization, the IP-internal topic or focus NP may precede or follow Tense-Aspect-Modality categories and sentential negation, but crucially, an IP-internal topic NP must precede and not follow the progressive aspect (*zheng*)*zai* or the aspectual/perfective negation (*hai*) *mei*, while an (IP-internal) focus NP always precedes *dou*, which is outside VoiceP; I take this to indicate that IP-internal topicalization and focalization has variable landing sites within the low IP area, but crucially must be structurally above the VoiceP (cf. Qu 1994; Shyu 1995; Ting 1995; Paul 2002, 2005; Kuo 2009).⁶

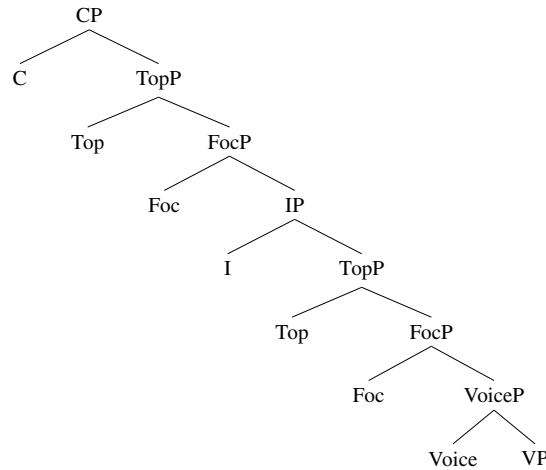
- (18) *IP-internal topic or focus NP precedes Tense-Aspect-Modality and sentential negation*

⁶(IP-internal) topicalization and focalization in Mandarin are subject to the so-called facilitator effects, which requires the movement to cross either a high facilitator outside the VoiceP (e.g., a Tense-Aspect-Modality category) or a low facilitator inside the VoiceP (e.g., a manner adverb) within its path. For more details, see Chen & Yip (2025).

- a. Lisi [(lian) **zhe-jian shi**]_i zuotian (dou) [_{VoiceP} (haohao)
 Lisi even this-CL matter yesterday DIST well
 diaocha-le —_i].
 investigate-PRF
 ‘Lisi, (even) this matter_i, yesterday investigated (it_i) (well).’
- b. Lisi [(lian) **zhe-jian shi**]_i (dou) keneng/keyi [_{VoiceP} (haohao)
 Lisi even this-CL matter DIST be.possible/be.able well
 diaocha —_i].
 investigate
 ‘Lisi, (even) this matter_i, is possible/is able to investigate (it_i) (well).’
- c. Lisi [(lian) **zhe-jian shi**]_i (dou) (zheng)zai/(hai) mei-you [_{VoiceP}
 Lisi even this-CL matter DIST PROG/yet NEG-have
 (haohao) diaocha —_i].
 well investigate
 ‘Lisi, (even) this matter_i, is investigating/has not investigated (it_i) (well).’
- d. *Lisi (zheng)zai/(hai) mei-you [**zhe-jian shi**]_i [_{VoiceP} (haohao)
 Lisi PROG/yet NEG-have this-CL matter well
 diaocha —_i].
 investigate
 INT: ‘Lisi, this matter_i, is investigating/has not investigated (it_i) (well).’

Hence, I adopt Paul’s (2005) proposal that in Mandarin, projections in the low IP area parallel those in the left periphery: IP-internal topicalization and focalization target IP-internal Spec, TopP and Spec, FocP in the low IP area, below the grammatical subject in Spec, IP; IP-external topicalization and focalization target IP-external Spec, TopP and Spec, FocP in the left periphery, above the grammatical subject in Spec, IP, as illustrated in (19).

(19) *Mandarin low IP area and left periphery*



Before proceeding, it is worth mentioning that both IP-internal and IP-external topicalization and focalization are subject to restrictions that call for [TOP] and [FOC] features. As seen in (20), the NP *mei-you shi* ‘nothing’ resists topicalization, whether it surfaces in an IP-internal or IP-external position, but does not resist passivization; this follows if NPs such as *mei-you shi* ‘nothing’ are incompatible with a [TOP] feature.

- (20) a. *Lisi [**mei-you shi**]_i hui diaocha ____i.
 Lisi not-have matter will investigate
 Int: ‘Lisi, nothing_i, will investigate (it_i).’
 b. *[**Mei-you shi**]_i, Lisi hui diaocha ____i.
 not-have matter Lisi will investigate
 Int: ‘Nothing_i, Lisi will investigate (it_i).’
 c. Cf. *Passivization*
 [**Mei-you shi**]_i hui bei Lisi diaocha ____i.
 not-have matter will BEI Lisi investigate
 ‘Nothing will be investigated by Lisi.’

As seen in (21), the wh-phrase *shei* ‘who’ cannot follow the focus NP; this follows if the focus NP has a [FOC] feature that induces intervention effects (see e.g., Li 2024).

- (21) a. *Lisi [**lian zhe-jian shi**]_i dou hui bi shei diaocha ___?
 Lisi even this-CL matter DIST will force who investigate
 Int: ‘Who is such that Lisi, even this matter_i, will force (him) to investigate (it_i)?’
 b. *[**Lian zhe-jian shi**]_i, Lisi dou hui bi shei diaocha ____i?
 even this-CL matter Lisi DIST will force who investigate
 Int: ‘Who is such that even this matter_i, Lisi will force (him) to investigate (it_i)?’

3.2 IP-internal topic and focus movement as successive-cyclic composite movement

It has long been noted that IP-internal topicalization and focalization can establish a long-distance dependency between the IP-internal topic or focus NP and a deeply embedded object gap across non-finite clause boundaries but not across a finite clause boundary (Qu 1994; Ting 1995; Shyu 1995; Kuo 2009):⁷

- (22) *IP-internal topicalization and focalization: Restricted long-distance dependency*

- a. *Long-distance dependency across non-finite boundaries*

⁷In (22), *mei-you ren* ‘nobody’ resists topicalization; hence, (22-b) cannot have a derivation where (i) the embedded object undergoes IP-external topicalization to precede the matrix subject, and (ii) the matrix subject undergoes further topicalization to precede the topicalized embedded object.

- Mei-you ren, [(lian) **zhe-jian shi**]_i, (dou) [_{VoiceP} bi Zhang [_{VoiceP}
not-have person even this-CL matter DIST force Zhang
jiao Wang [_{VoiceP} diaocha —_i]].
ask Wang investigate
'Nobody, (even) this matter_i, forced Zhang to ask Wang to investigate
(it_i).'
- b. *No long-distance dependency across finite boundary*
*Mei-you ren, [(lian) **zhe-jian shi**]_i, (dou) xiangxin/zhidao [_{CP}
no-have person even this-CL matter DIST believe/know
Zhang hui diaocha —_i].
Zhang will investigate
INT: 'Nobody, (even) this matter_i, believes/knows that Zhang will inves-
tigate (it_i).'

In addition, (long-distance) IP-internal topicalization and focalization exhibit prop-
erties of A-move-ment, creating new antecedents for anaphor binding and not subject
to weak crossover or Principle C reconstruction (Qu 1994; Ting 1995; Shyu 1995; Kuo
2009).

- (23) a. *IP-internal topicalization and focalization: New antecedents for anaphor
binding*
Zhang, [(lian) **zhe-ge ren**]_i, (dou) [_{VoiceP} bi ta-ziji_i-de pengyou
Zhang even that-CL person DIST force 3SG-self's friend
[_{VoiceP} ma-guo —_i]].
scold-EXP
'Zhang, (even) this person_i, once forced his_i friend to scold (him_i).'
- b. *IP-internal topicalization (of quantifier phrase): No weak crossover*
Zhang, [**mei-ge ren**]_i, dou [_{VoiceP} bi ta_i-de pengyou [_{VoiceP}
Zhang every-CL person DIST force 3SG's friend
ma-guo —_i]]?
scold-EXP
Lit. 'Zhang, every person_i, once forced his_i friend to scold (him_i)?'
- c. *IP-internal focalization: No weak crossover*
Zhang, [(lian) **Li**]_i, dou [_{VoiceP} bi ta_i-de pengyou [_{VoiceP} ma-guo
Zhang even Li DIST force 3SG's friend scold-EXP
—_i]]?

Lit. 'Zhang, even Li_i, once forced his_i friend to scold (him_i).'
- d. *IP-internal topicalization and focalization: No Principle C reconstruc-
tion*
Zhang, [(lian) **Li-de pengyou**]_j, (dou) [_{VoiceP} bi ta_i [_{VoiceP}
Zhang even Li's friend DIST force 3SG

ma-guo —_j]].
 scold-EXP
 ‘Zhang, (even) Li_i’s friend_j, once forced him_i to scold (him_j).’

In contrast to existing work that analyzes IP-internal topicalization and focalization as instances of A-movement, thereby overlooking their lack of A-minimality/locality effects and the presence of information-structural effects, I propose that IP-internal topicalization and focalization involve successive-cyclic composite A/ $\bar{A}$ -movement via Spec, VoicePs, terminating at IP-internal Spec, TopP, and Spec, FocP, respectively, as illustrated in (24). This derivation necessitates that both the Voice head, as a phase head, and the IP-internal Top and Foc heads host the composite probes $[\phi + \text{TOP}]$ and $[\phi + \text{FOC}]$, and it follows that this composite A/ $\bar{A}$ -movement should exhibit mixed A/ $\bar{A}$ -properties.

- (24) a. *Mandarin: IP-internal topicalization as successive-cyclic composite movement*

$$[_{\text{IP}} \text{Subj}_{i[\phi]} \text{Infl}_{[\phi]} [_{\text{TopP}} \text{Obj}_{j[\phi],[\text{TOP}]} \text{Top}_{[\phi+\text{TOP}]} \dots [_{\text{VoiceP}} t_i t_j \text{Voice}_{[\phi+\text{TOP}]} \dots t_j (\dots)]]]$$

$$\underbrace{\text{Obj}_{j[\phi],[\text{TOP}]} \text{Top}_{[\phi+\text{TOP}]} \dots t_i t_j}_{\text{A}/\bar{\text{A}}\text{-movement}} \underbrace{\text{Voice}_{[\phi+\text{TOP}]} \dots t_j (\dots)}_{\text{A}/\bar{\text{A}}\text{-movement}}$$
- b. *Mandarin: IP-internal focalization as successive-cyclic composite movement*

$$[_{\text{IP}} \text{Subj}_{i[\phi]} \text{Infl}_{[\phi]} [_{\text{FocP}} \text{Obj}_{j[\phi],[\text{FOC}]} \text{Foc}_{[\phi+\text{FOC}]} \dots [_{\text{VoiceP}} t_i t_j \text{Voice}_{[\phi+\text{FOC}]} \dots t_j (\dots)]]]$$

$$\underbrace{\text{Obj}_{j[\phi],[\text{FOC}]} \text{Foc}_{[\phi+\text{FOC}]} \dots t_i t_j}_{\text{A}/\bar{\text{A}}\text{-movement}} \underbrace{\text{Voice}_{[\phi+\text{FOC}]} \dots t_j (\dots)}_{\text{A}/\bar{\text{A}}\text{-movement}}$$

Furthermore, it is important that the Mandarin C head, also as a phase head, only hosts *pure $\bar{A}$ -probes* (e.g., $[\text{TOP}]$, $[\text{FOC}]$) and not the composite probes $[\phi + \text{TOP}]$ and $[\phi + \text{FOC}]$.⁸ Consequently, $\bar{A}$ -movement to Spec, CP cannot feed composite A/ $\bar{A}$ -movement to the IP-internal Spec, TopP and Spec, FocP, due to the ban on improper composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement (see e.g., Longenbaugh 2017; Chen 2023).

- (25) *Improper composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement*
- a.
$$*[_{\text{IP}} \text{Subj}_{i[\phi]} \text{Infl}_{[\phi]} [_{\text{TopP}} \text{Obj}_{j[\phi],[\text{TOP}]} \text{Top}_{[\phi+\text{TOP}]} \dots [_{\text{VoiceP}} t_i t_j \dots [_{\text{CP}} t_j \text{C}_{[\text{TOP}]} \dots t_j (\dots)]]]]$$

$$\underbrace{\text{Obj}_{j[\phi],[\text{TOP}]} \text{Top}_{[\phi+\text{TOP}]} \dots t_i t_j}_{\text{A}/\bar{\text{A}}\text{-movement}} \dots \underbrace{t_j \text{C}_{[\text{TOP}]} \dots t_j (\dots)}_{\text{A-movement}}$$
- b.
$$*[_{\text{IP}} \text{Subj}_{i[\phi]} \text{Infl}_{[\phi]} [_{\text{FocP}} \text{Obj}_{j[\phi],[\text{FOC}]} \text{Foc}_{[\phi+\text{FOC}]} \dots [_{\text{VoiceP}} t_i t_j \dots [_{\text{CP}} t_j \text{C}_{[\text{FOC}]} \dots t_j (\dots)]]]]$$

$$\underbrace{\text{Obj}_{j[\phi],[\text{FOC}]} \text{Foc}_{[\phi+\text{FOC}]} \dots t_i t_j}_{\text{A}/\bar{\text{A}}\text{-movement}} \dots \underbrace{t_j \text{C}_{[\text{FOC}]} \dots t_j (\dots)}_{\text{A-movement}}$$

⁸In Chen (2023), it is argued that the Mandarin C head also hosts a *pure ϕ -probe*, which is active in the case of hyper-raising to subject via Spec, CP. Hence, both an [A]-feature and [$\bar{A}$]-features are present on the Mandarin C head, but, crucially, they probe separately and independently.

3.3 IP-external topic and focus movement via intermediate composite movement

Unlike IP-internal topicalization and focalization, which are clause-bound, and like typical $\bar{A}$ -movement, IP-external topicalization and focalization can establish a long-distance dependency between the IP-external topic or focus NP and a deeply embedded object gap across a finite clause boundary:

- (26) *IP-external topicalization and focalization: Long-distance dependency across finite boundary*

[(Lian) **zhe-jian shi**]_i, Li (dou) xiangxin/zhidao [_{CP} Zhang hui diaocha
even this-CL matter Li DIST believe/know Zhang will investigate
—_i].

‘(Even) this matter_i, Li believes/knows that Zhang will investigate (it_i).’

Also unlike IP-internal topicalization and focalization, IP-external topicalization and focalization do not create new antecedents for anaphor binding, regardless of whether the anaphor is present between the IP-external topic or focus NP and the matrix VoiceP, as in (27-a), or between the embedded CP and the embedded VoiceP, as in (27-b), or between the matrix VoiceP and the embedded VoiceP, as in (27-c) (cf. Qu 1994; Ting 1995; Shyu 1995; Kuo 2009).

- (27) *IP-external topicalization and focalization: No new antecedents for anaphor binding*

- a. ... *anaphor between topic/focus and matrix VoiceP*

*[(Lian) **Li**]_i, ta-ziji_i-de pengyou (dou) [_{VoiceP} ma-guo —_i].
even Li 3SG-self’s friend DIST scold-EXP
INT: ‘(Even) Li_i, his_i friend once scolded (him_i).’

- b. ... *anaphor between embedded CP and embedded VoiceP*

*[(Lian) **Li**]_i, Zhang (dou) xiangxin/zhidao [_{CP} ta-ziji_i-de pengyou
even Li Zhang DIST believe/know 3SG-self’s friend
[_{VoiceP} ma-guo —_i]].
scold-EXP
INT: ‘(Even) Li_i, Zhang believes/knows that his_i friend once scolded (him_i).’

- c. ... *anaphor between matrix VoiceP and embedded VoiceP*

*[(Lian) **Li**]_i, Zhang (dou) [_{VoiceP} bi ta-ziji_i-de pengyou [_{VoiceP}
even Li Zhang DIST force 3SG-self’s friend
ma-guo —_i]].
scold-EXP
INT: ‘(Even) Li_i, Zhang once forced his_i friend to scold (him_i).’

More interestingly, whether IP-external topicalization and focalization are subject to weak crossover and Principle C reconstruction depends on the position of the pronoun co-indexed with the IP-external topic or focus NP in the given configurations (cf. Qu 1994).⁹ Specifically, IP-external topicalization and focalization are subject to weak crossover, when the co-indexed pronoun is outside the VoiceP phase(s), as in (28-a), (28-b) and (29-a), (29-b) (cf. Qu 1994; Ting 1995; Shyu 1995; Kuo 2009), but is immune to weak crossover, when the co-indexed pronoun is inside the VoiceP phase(s), as in (28-c) and (29-c) (cf. Qu 1994).

(28) *IP-external topicalization (of quantifier phrase): (No) weak crossover*

- a. ... co-indexed pronoun **between topic and matrix VoiceP**
 *[Mei-ge ren]_i, ta_i-de pengyou (dou) [_{VoiceP} ma-guo _____i]].
 every-CL person 3SG's friend DIST scold-EXP
 INT: 'Every person_i, his_i friend once scolded (him_i).'
- b. ... co-indexed pronoun **between embedded CP and embedded VoiceP**
 *[Mei-ge ren]_i, Zhang (dou) xiangxin/zhidao [_{CP} ta_i-de pengyou [_{VoiceP} every-CL person Zhang DIST believe/know 3SG's friend ma-guo _____i]].
 scold-EXP
 INT: 'Every person_i, Zhang believes/knows that his_i friend once scolded (him_i).'
- c. ... co-indexed pronoun **between matrix VoiceP and embedded VoiceP**
 [Mei-ge ren]_i, Zhang (dou) [_{VoiceP} bi ta_i-de pengyou [_{VoiceP} every-CL person Zhang DIST force 3SG's friend ma-guo _____i]].
 scold-EXP
 Lit. 'Every person_i, Zhang once forced his_i friend to scold (him_i).'

(29) *IP-external focalization: (No) weak crossover*

- a. ... co-indexed pronoun **between focus and matrix VoiceP**
 *[Lian Li]_i, ta_i-de pengyou (dou) [_{VoiceP} ma-guo _____i]].
 even Li 3SG's friend DIST scold-EXP
 INT: 'Even Li_i, his_i friend once scolded (him_i).'
- b. ... co-indexed pronoun **between embedded CP and embedded VoiceP**
 *[Lian Li]_i, Zhang (dou) xiangxin/zhidao [_{CP} ta_i-de pengyou [_{VoiceP} even Li Zhang DIST believe/know 3SG's friend ma-guo _____i]].
 scold-EXP

⁹Qu (1994) fails to identify the link between the presence or absence of weak crossover and Principle C reconstruction and the position of the co-indexed pronoun and simply concludes that IP-external topicalization and focalization may be instances of either A-movement, when they are immune to weak crossover and Principle C reconstruction, or $\bar{A}$ -movement, when they are subject to weak crossover and Principle C reconstruction. This is a logical flaw: if IP-external topicalization and focalization can be instances of A-movement, then any ungrammaticality due to weak crossover or Principle C reconstruction should be circumventable, contrary to fact.

INT: ‘Even Li_i, Zhang believes/knows that his_i friend once scolded (him_i).’

- c. ... *co-indexed pronoun between matrix VoiceP and embedded VoiceP*
 [Lian **Li**]_i, Zhang (dou) [_{VoiceP} bi ta_i-de pengyou [_{VoiceP} ma-guo
 even Li Zhang DIST force 3SG’s friend scold-EXP
 —_i]].

Lit. ‘Even Li_i, Zhang once forced his_i friend to scold (him_i).’

Similarly, IP-external topicalization and focalization show (weak) Principle C reconstruction effects, when the co-indexed pronoun is outside the VoiceP phase(s), as in (30-a) and (30-b),¹⁰ but show no Principle C reconstruction effects, when the co-indexed pronoun is inside the VoiceP phase(s), as in (30-c) (cf. Qu 1994).

(30) *IP-external topicalization and focalization: (No) Principle C reconstruction*¹¹

- a. ... *co-indexed pronoun between topic/focus and matrix VoiceP*
 */?? [(Lian) **zhe-ge ren_i-de pengyou**]_j, ta_i (dou) [_{VoiceP} ma-guo
 even this-CL person’s friend 3SG DIST scold-EXP
 —_j]].

INT: ‘(Even) this person_i’s friend_j, he_i once scolded (him_j).’

- b. ... *co-indexed pronoun between embedded CP and embedded VoiceP*
 */? [(Lian) **zhe-ge ren_i-de pengyou**]_j, Zhang (dou) xiangxin/zhidao
 even this-CL person’s friend Zhang DIST believe/know
 [_{CP} ta_i [_{VoiceP} ma-guo —_j]].
 3SG scold-EXP

‘(Even) this person_i’s friend_j, Zhang believes/knows that he_i once scolded (him_j).’

- c. ... *co-indexed pronoun between matrix VoiceP and embedded VoiceP*
 [(Lian) **zhe-ge ren_i-de pengyou**]_j, Zhang (dou) [_{VoiceP} bi ta_i [_{VoiceP}
 even this-CL person’s friend Zhang DIST force 3SG
 ma-guo —_j]].
 scold-EXP

‘(Even) this person_i’s friend_j, Zhang once forced him_i to scold (him_j).’

I propose that IP-external topicalization and focalization in Mandarin, which otherwise involve typical successive-cyclic $\bar{A}$ -movement via VoiceP and CP phase edges, can also involve intermediate steps of composite A/ $\bar{A}$ -movement to Spec, VoicePs, followed by a terminating step of $\bar{A}$ -movement to IP-external Spec, TopP and Spec, FocP.

¹⁰In (30-a), Principle C reconstruction effect is weak with topicalization and stronger with focalization. In (30-b), Principle C reconstruction effect is further weakened with topicalization (cf. Huang 1993: ex. 54c), which might indicate that cross-clausal reconstruction for Principle C is optional with topicalization.

The intermediate composite A/ $\bar{A}$ -movement to Spec, VoiceP is associated with mixed A/ $\bar{A}$ -properties (e.g., the lack of weak crossover/Principle C reconstruction), while the terminating $\bar{A}$ -movement to IP-external Spec, TopP and Spec, FocP is associated with $\bar{A}$ -properties (e.g., weak crossover/Principle C reconstruction).

- (31) a. *Mandarin: IP-external topicalization via intermediate composite movement*

[_{TopP} **Obj**_{i[ϕ],[TOP]} Top_[TOP] [_{IP} Subj_{j[ϕ]} Infl_[ϕ] ... [_{VoiceP} t_i t_j Voice_[ϕ +TOP] ... t_i (...)]]]

$\xrightarrow{\bar{A}\text{-movement}}$
 $\xrightarrow{A/\bar{A}\text{-movement}}$

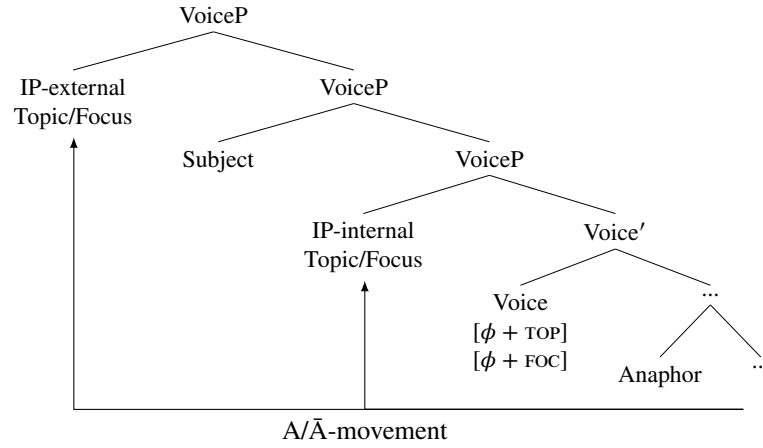
- b. *Mandarin: IP-external focalization via intermediate composite movement*

[_{FocP} **Obj**_{i[ϕ],[FOC]} Foc_[FOC] [_{IP} Subj_{j[ϕ]} Infl_[ϕ] ... [_{VoiceP} t_i t_j Voice_[ϕ +FOC] ... t_i (...)]]]

$\xrightarrow{\bar{A}\text{-movement}}$
 $\xrightarrow{A/\bar{A}\text{-movement}}$

Recall that IP-internal topicalization and focalization can, while IP-external topicalization and focalization cannot, create new antecedents for anaphor binding, even when the anaphor is inside a VoiceP phase. I propose that at the edge of a VoiceP phase, the external argument is the SUBJECT that delimits the binding domain for anaphors. Furthermore, IP-internal topicalization and focalization stop off at an *inner* Spec, VoiceP, below the thematic position of the external argument, while IP-external topicalization and focalization stop off at an *outer* Spec, VoiceP, above the thematic position of the external argument; in this way, the linear orderings of phrases established by phase-by-phase spell-out are consistent throughout a derivation (see e.g., Fox & Pesetsky 2005; Davis 2020). Consequently, in the former case, the topic or focus NP is *inside* the binding domain for anaphors that lack a more locally accessible SUBJECT, while in the latter case, the topic or focus NP is *outside* the binding domain for anaphors that are bound by the external argument in Spec, VoiceP/its thematic position, the more locally accessible SUBJECT.

- (32) *Topicalization/Focalization and anaphor binding at VoiceP phase*



3.4 A note on Principle A reconstruction

In Mandarin, there is also a contrast between IP-internal and IP-external topicalization and focalization in terms of Principle A reconstruction effects (see e.g., Qu 1994; Ting 1995; Shyu 1995; Kuo 2009):

- (33) a. *IP-internal topicalization and focalization: no Principle A reconstruction*
 Zhangsan_i [(lian) **ta-ziji_{i/*j}-de pengyou**] (dou) bi Lisi_j ma-guo ____.
 Zhangsan even 3SG-self's friend DIST force Lisi scold-EXP
 'Zhangsan_i, (even) his_{i/*j} friend, forced Lisi_j to scold (him).'
 b. *IP-external topicalization and focalization: Principle A reconstruction*
 [(Lian) **ta-ziji_{i/j}-de pengyou**], Zhangsan_i (dou) bi Lisi_j ma-guo ____.
 even 3SG-self's friend Zhangsan DIST force Lisi scold-EXP
 '(Even) his_{i/j} friend, Zhangsan_i forced Lisi_j to scold (him).'

Cross-linguistically, there is variation in whether A-movement shows Principle A reconstruction effects. In English, both A-movement and $\bar{A}$ -movement show Principle A reconstruction effects (Belletti & Rizzi 1988; Pesetsky 2013), while in Dutch, only $\bar{A}$ -movement shows Principle A reconstruction effects (see e.g., Neeleman & Van De Koot 2010).

I propose that Mandarin is unlike English and like Dutch in that only $\bar{A}$ -movement shows Principle A reconstruction effects. The contrast in (33) follows because IP-internal topicalization and focalization involve successive-cyclic composite A/ $\bar{A}$ -movement, which does not show Principle A reconstruction effects, while IP-external topicalization and focalization can involve typical successive-cyclic $\bar{A}$ -movement, which shows Principle A reconstruction effects.¹²

¹²Given the above proposal that the external argument in Spec, VoiceP/its thematic position can bind an

3.5 A note on VP and CP topicalization and focalization

In addition to an NP, a VP and a CP can also undergo IP-internal and IP-external topicalization and focalization, as seen in (34) and (35) (see e.g., Huang 1993; Paul 2002, 2005; a.o.).^{13,14}

- (34) a. *IP-internal VP topicalization/focalization*
 Lisi [(lian) [_{VP} **diaocha zhe-jian shi**]]_i (dou) bu hui ___i.
 Lisi even investigate this-CL matter DIST not will
 ‘Lisi, (even) investigate this matter, will not (do it).’
- b. *IP-external VP topicalization/focalization*
 [(Lian) [_{VP} **diaocha zhe-jian shi**]]_i, Lisi (dou) bu hui ___i.
 even investigate this-CL matter Lisi DIST not will
 ‘(Even) investigate this matter, Lisi will not (do it).’
- (35) a. *IP-internal CP topicalization/focalization*
 Lisi [(lian) [_{CP} **Zhangsan diaocha-le zhe-jian shi**]]_i (dou) bu
 Lisi even Zhangsan investigate-PRF this-CL matter DIST not
 xiangxin/zhidao ___i.
 believe/know
 ‘Lisi, (even) that Zhangsan investigated this matter, does not believe/know.’
- b. *IP-external CP topicalization/focalization*
 [(Lian) [_{CP} **Zhangsan diaocha-le zhe-jian shi**]]_i Lisi (dou) bu
 even Zhangsan investigate-PRF this-CL matter Lisi DIST not
 xiangxin/zhidao ___i.
 believe/know
 ‘(Even) that Zhangsan investigated this matter, Lisi does not believe/know.’

Like IP-internal NP topicalization and focalization, IP-internal VP and CP topicalization and focalization do not show reconstruction effects for Principle C, as seen in (36).

- (36) a. *IP-internal VP topicalization and focalization: no Principle C reconstruction*
 Zhangsan [(lian) [_{VP} **ma Lisi-de pengyou**]]_j (dou) bi-guo ta_i ___j.
 Zhangsan even scold Lisi’s friend DIST force-EXP 3SG
 ‘Zhangsan, even scold Lisi_i’s friend, once forced him_i to (do it).’

anaphor before its A-movement to Spec, IP, it remains puzzling as to why an anaphor cannot be bound at its thematic position before (A-movement and) composite A/Ā-movement. I leave this puzzle and an account of the cross-linguistic variation on Principle A reconstruction effects to future research.

¹³Huang (1993) has provided convincing evidence that in apparent VP fronting, the fronted phrase is in fact structurally as large as a VoiceP, which contains the subject trace in Spec, VoiceP. This detail is irrelevant to the discussion here.

¹⁴As noted previously in footnote 5, a fronted VP or CP to an IP-internal position requires a contrastive topic or focus reading.

- b. *IP-internal CP topicalization and focalization: no Principle C reconstruction*

Zhangsan [(Lian) [_{CP} **Lisi_i-de pengyou si-le**]]_j (dou) mei-you gaosu
 Zhangsan even Lisi's friend die-PRF DIST not-have tell
 ta_i —_j.
 3SG
 'Zhangsan, even that Lisi_i's friend died, did not tell him_i.'

Similarly, like IP-external NP topicalization and focalization, IP-external VP and CP topicalization and focalization show (weak) Principle C reconstruction effects, when the co-indexed pronoun is outside the VoiceP phase(s), as in (37-a), (37-b) and (38-a), (38-b),¹⁵ but show no Principle C reconstruction effects, when the co-indexed pronoun is inside the VoiceP phase(s), as in (37-c) and (38-c).

- (37) *IP-external VP topicalization and focalization: (no) Principle C reconstruction*

- a. ... co-indexed pronoun *between topic and matrix VoiceP*
 */?? [(Lian) [_{VP} **ma Lisi_i-de pengyou**]]_j, ta_i (dou) bu hui —_j.
 even scold Lisi's friend 3SG DIST not will
 Int: '(Even) scold Lisi_i's friend, he_i will not (do it).'
- b. ... co-indexed pronoun *between embedded CP and embedded VoiceP*
 */? [(Lian) [_{VP} **ma Lisi_i-de pengyou**]]_j Zhangsan (dou)
 even scold Lisi's friend Zhangsan DIST
 xiangxin/zhidao [_{CP} ta_i bu hui —_j].
 believe/know 3SG not will
 Int: '(Even) scold Lisi_i's friend, Zhangsan believes/knows that he_i will not (do it).'
- c. ... co-indexed pronoun *between matrix VoiceP and embedded VoiceP*
 [(Lian) [_{VP} **ma Lisi_i-de pengyou**]]_j, Zhangsan (dou) bi-guo ta_i
 even Lisi's friend die-PRF Zhangsan DIST force-EXP 3SG
 —_j.
 '(Even) scold Lisi_i's friend, Zhangsan once forced him_i to (do it).'

- (38) *IP-external CP topicalization and focalization: (no) Principle C reconstruction*

- a. ... co-indexed pronoun *between topic and matrix VoiceP*
 */?? [(Lian) [_{CP} **Lisi_i-de pengyou si-le**]]_j, ta_i (dou) bu
 even Lisi's friend die-PRF 3SG DIST not
 xiangxin/zhidao —_j.
 believe/know

¹⁵ Again, Principle C reconstruction effect is weak with topicalization and stronger with focalization.

- ‘(Even) that Lisi_i’s friend died, he_i does not believe/know.’
- b. ... *co-indexed pronoun between embedded CP and embedded VoiceP*
 */? [(Lian) [_{CP} **Lisi_i-de pengyou si-le**]]_j Zhangsan (dou)
 even Lisi’s friend die-PRF Zhangsan DIST
 xiangxin/zhidao [_{CP} ta_i bu xiangxin/zhidao ___j].
 believe/know 3SG not believe/know
 ‘(Even) that Lisi_i’s friend died, Zhangsan believes/knows that he_i does not believe/know.’
- c. ... *co-indexed pronoun between matrix VoiceP and embedded VoiceP*
 [(Lian) [_{CP} **Lisi_i-de pengyou si-le**]]_j, Zhangsan (dou) mei-you gaosu
 even Lisi’s friend die-PRF Zhangsan DIST not-have tell
 ta_i ___j.
 3SG
 ‘(Even) that Lisi_i’s friend died, Zhangsan did not tell him_i.’

This is not surprising, given that, like an NP, a VP and a CP can also be a grammatical subject, as seen in (39), and undergo passivization, as seen in (40). All that needs to be assumed is that a VP and a CP can also have an [A]-feature in addition to an [Ā]-feature, allowing them to be targeted by a pure A-probe, a composite A/=A-probe, or a pure Ā-probe.

- (39) a. [_{VP} **Diaocha zhe-jian shi**] hen rongyi.
 investigate this-CL matter DEG easy
 ‘Investigate this matter is easy.’
 b. [_{CP} **Zhangsan mousha-le Lisi**] shi zhen-de.
 Zhangsan murder-PRF Lisi COP true
 ‘That Zhangsan murdered Lisi is true.’
- (40) a. [_{VP} **Kaobei zhe-fen zilia**]_i bei Lisi fanfu changshi-guo ___i.
 copy this-CL document BEI Lisi repeatedly try-EXP
 ‘Copy this document was repeatedly tried by Lisi.’
 b. [_{CP} **Zhangsan mousha-le Lisi**] hui bei suoyou-ren xiangxin/zhidao
 Zhangsan murder-PRF Lisi will BEI every-person believe/know
 ___i
 ‘That Zhangsan murdered Lisi will be believed/known by everyone.’

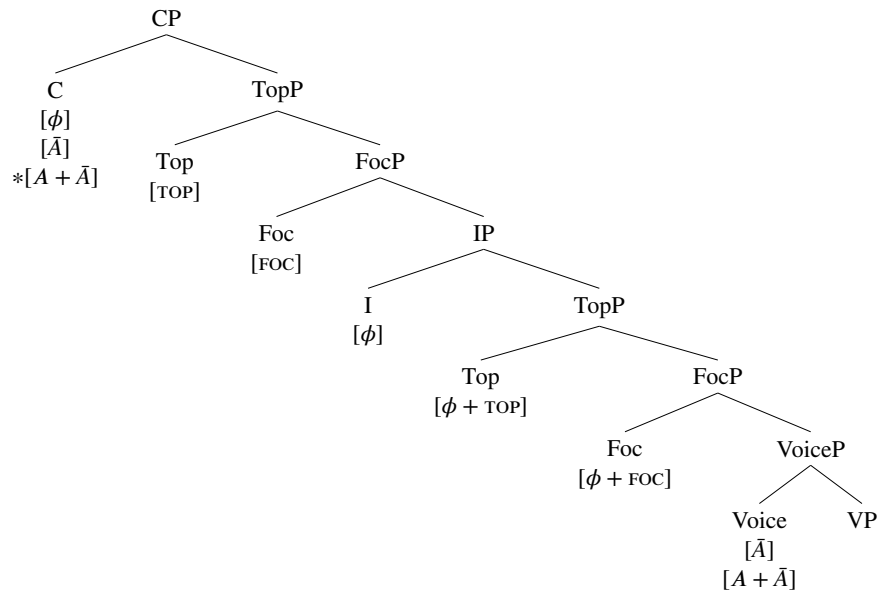
4 Conclusion

To summarize, I have argued that composite probing by the composite probe [A+Ā] is generally observed in Mandarin, in the sense that multiple heads projected in the low IP area host the composite probe [A+Ā], and that the Voice head, as a phase head,

generally hosts the composite probe $[A + \bar{A}]$ for purposes of successive-cyclic movement. The evidence has come from two types of topicalization and focalization in Mandarin, which exhibit mixed $A/\bar{A}$ -properties. Specifically, I have argued that IP-internal topicalization and focalization involve successive-cyclic composite $A/\bar{A}$ -movement via Spec, VoicePs, which terminate at IP-internal Spec, TopP and Spec, FocP, respectively, while IP-external topicalization and focalization can involve intermediate steps of composite $A/\bar{A}$ -movement to Spec, VoicePs, followed by a terminating step of $\bar{A}$ -movement to IP-external Spec, TopP and Spec, FocP, respectively.

The proposed analyses lead to the distribution of A -, $\bar{A}$ -, and composite probes in Mandarin low IP area and left periphery in (41). The IP-internal Top head and Foc head host composite $A/\bar{A}$ -probes, $[\phi + \text{TOP}]$ and $[\phi + \text{FOC}]$ respectively, which trigger IP-internal topicalization and focalization, while the IP-external Top head and Foc head host pure $\bar{A}$ -probes, $[\text{TOP}]$ and $[\text{FOC}]$ respectively, which trigger IP-external topicalization and focalization. The Voice head, as a phase head, hosts both composite $A/\bar{A}$ -probes and pure $\bar{A}$ -probes; by contrast, the C head, also as a phase head, does not host composite $A/\bar{A}$ -probes.

- (41) *Distribution of A -, $\bar{A}$ -, and composite probes in Mandarin low IP area and left periphery*



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